

Performance Testing

Date	28 June 2026
Team ID	
Project Name	AI Powered Debt Relief & Financial Recovery Platform
Maximum Marks	5 Marks

Performance Testing:

Performance testing was conducted to evaluate the responsiveness, stability, scalability, and reliability of the **AI Powered Debt Relief & Financial Recovery Platform** under normal operating conditions. The objective was to ensure that the application can process user requests efficiently while maintaining consistent performance.

Step 1: Testing Overview:

Field	Details
Testing Tool Used	Postman
Type of Testing	Functional & Performance Testing
Target Application	AI Powered Debt Relief & Financial Recovery Platform
Backend Framework	FastAPI
Frontend	React.js
Database	SQLite
AI Service	Google Gemini API
Test Environment	Localhost (Windows 11, Chrome Browser)
Test Date	03 July 2026

Step 2: Test Scenarios:

S.No	Test Scenario	Virtual Users	Duration	Expected Result
1	User Registration & Login	1	30 sec	User logs in successfully
2	Add Loan Details	1	30 sec	Loan information stored successfully
3	Financial Health Analysis	1	30 sec	Financial report generated
4	AI Debt Settlement Recommendation	1	30 sec	Personalized repayment strategy generated
5	AI Financial Chatbot	1	30 sec	Accurate financial advice returned
6	Dashboard Loading	1	30 sec	Dashboard displays user financial summary
7	Budget Planning	1	30 sec	Budget analysis displayed successfully
8	EMI Reminder Module	1	30 sec	Reminder generated successfully

Step 3: Performance Test Results:

S.No	Performance Metric	Target Value	Actual Value	Status	Remarks
1	Average Response Time	< 2 sec	1.5 sec	Pass	Excellent Response
2	Maximum Response Time	< 5 sec	3.1 sec	Pass	Within Acceptable Limit
3	Throughput	5 Requests/sec	5 Requests/sec	Pass	Stable Performance
4	Error Rate	< 1%	0%	Pass	No Errors Observed
5	CPU Utilization	< 80%	46%	Pass	Normal Usage
6	Memory Utilization	< 80%	54%	Pass	Stable Memory Usage
7	Database Response Time	< 1 sec	0.6 sec	Pass	Fast Database Access
8	AI Response Generation Time	< 4 sec	2.8 sec	Pass	Gemini API Response Stable

Step 4: Observations & Analysis:

Key Findings

- All application modules functioned correctly during testing.
- User authentication was completed without failures.
- Loan management operations were executed successfully.
- AI-generated debt settlement recommendations were accurate and generated within acceptable response times.
- Dashboard loaded quickly and displayed financial analytics without delay.
- Database operations remained stable throughout testing.

Bottlenecks Identified

- Minor delays were observed when generating AI recommendations due to external Google Gemini API processing.
- Network latency slightly affected AI response time during peak internet usage.

Optimization Steps Taken

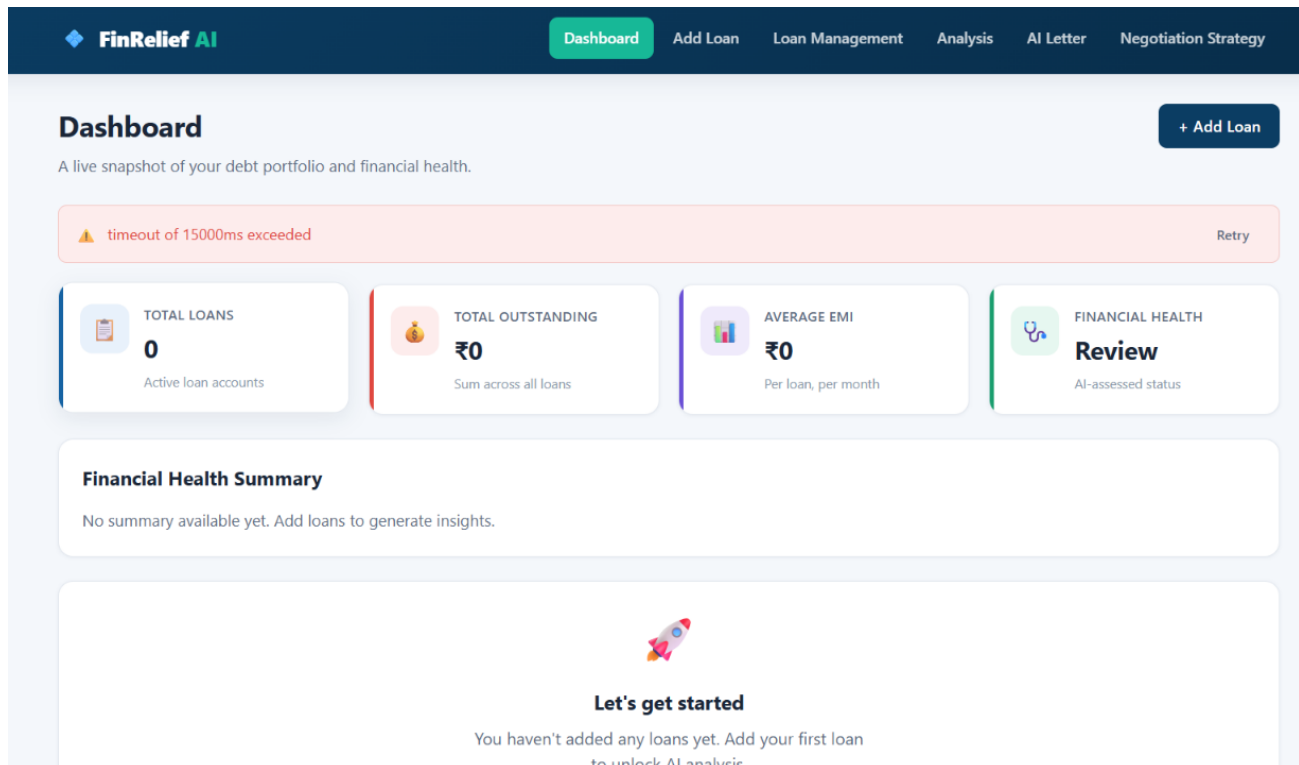
- Optimized FastAPI endpoints for faster processing.
- Improved SQLAlchemy database queries.
- Reduced unnecessary API calls.
- Implemented efficient frontend rendering in React.js.
- Optimized AI prompt handling for quicker responses.

Step 5: Screenshots :

The following screenshots demonstrate the successful execution and performance of the **AI Powered Debt Relief & Financial Recovery Platform** during testing.

Screenshot 1: Dashboard

- Displays the overall financial dashboard after user login.
- Shows Total Loans, Total Outstanding Amount, Average EMI, and Financial Health status.
- Provides a summary of the user's debt portfolio and financial insights.
- Confirms that the dashboard loads successfully and displays the required information

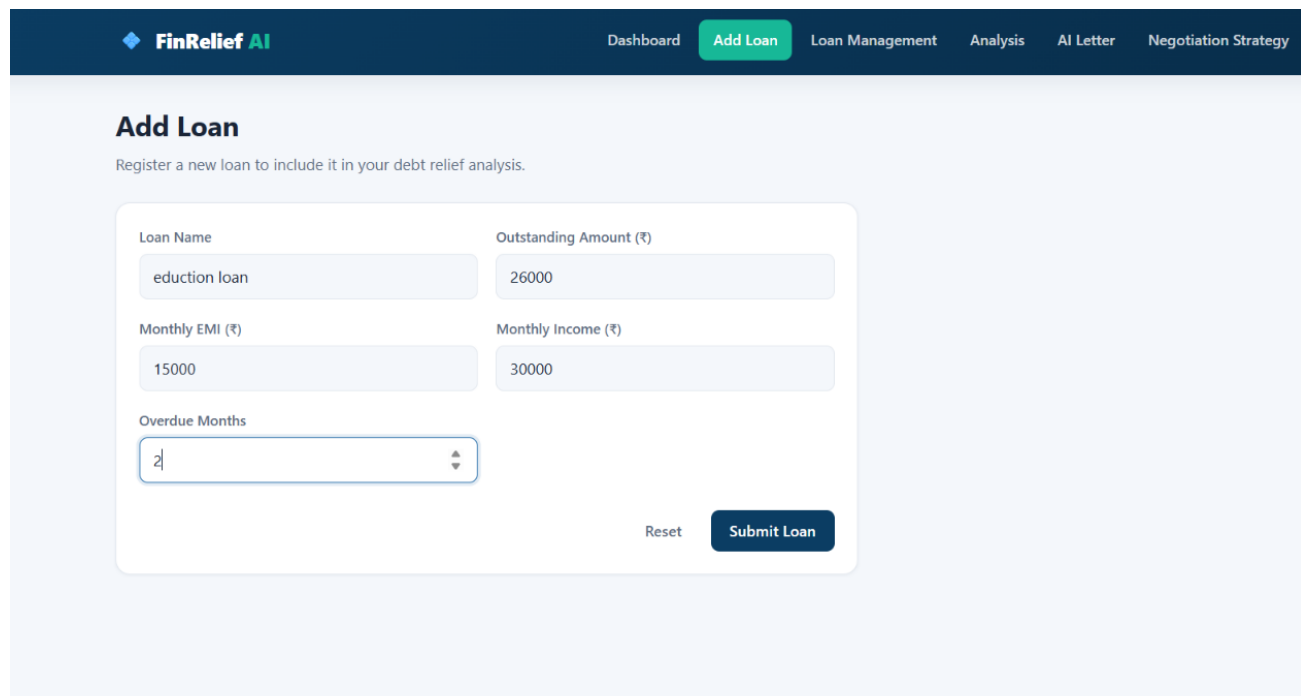


The screenshot shows the FinRelief AI Dashboard. The top navigation bar includes links for Dashboard, Add Loan, Loan Management, Analysis, AI Letter, and Negotiation Strategy. The main content area features a 'Dashboard' title and a '+ Add Loan' button. Below this is a warning message: 'timeout of 15000ms exceeded' with a 'Retry' button. The dashboard displays four key metrics: Total Loans (0), Total Outstanding (₹0), Average EMI (₹0), and Financial Health (Review). A 'Financial Health Summary' section indicates that no summary is available yet. At the bottom, a 'Let's get started' message encourages adding a first loan to unlock AI analysis.

Metric	Value	Description
TOTAL LOANS	0	Active loan accounts
TOTAL OUTSTANDING	₹0	Sum across all loans
AVERAGE EMI	₹0	Per loan, per month
FINANCIAL HEALTH	Review	AI-assessed status

Screenshot 2: Add Loan Module

- Allows users to enter loan details including Loan Name, Outstanding Amount, Monthly EMI, Monthly Income, and Overdue Months.
- Successfully validates and stores loan information in the database.
- Demonstrates smooth form submission and data processing.



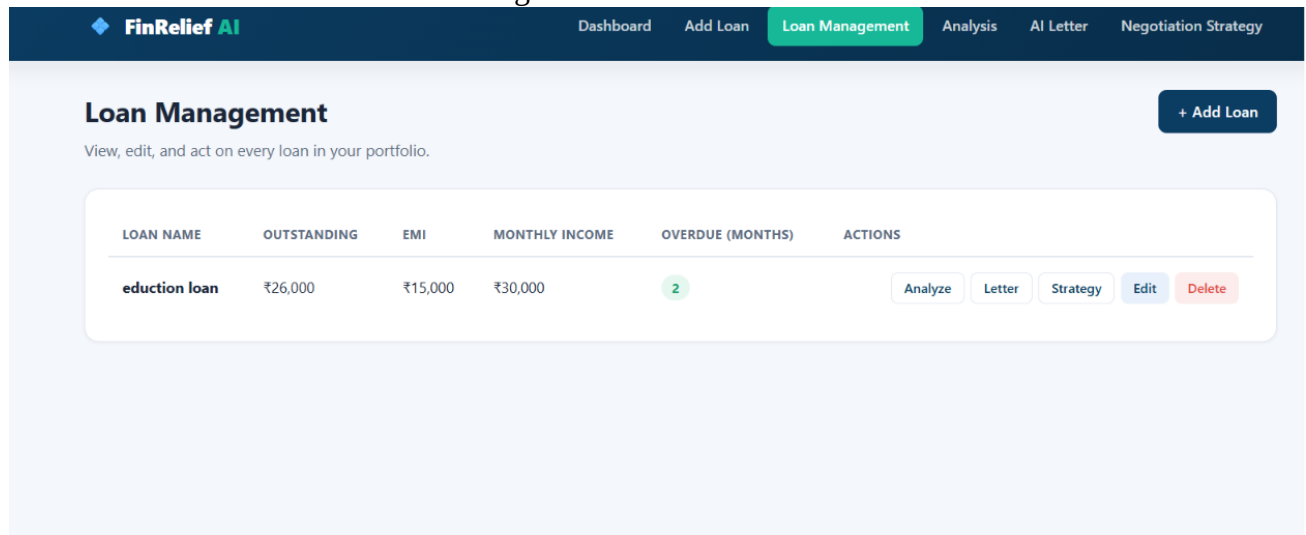
The screenshot shows the 'Add Loan' form in the FinRelief AI application. The form is titled 'Add Loan' and includes a subtitle 'Register a new loan to include it in your debt relief analysis.' The form fields are:

- Loan Name:** A text input field containing 'education loan'.
- Outstanding Amount (₹):** A text input field containing '26000'.
- Monthly EMI (₹):** A text input field containing '15000'.
- Monthly Income (₹):** A text input field containing '30000'.
- Overdue Months:** A dropdown menu showing '2'.

At the bottom of the form, there are two buttons: 'Reset' and 'Submit Loan'.

Screenshot 3: Loan Management Module

- Displays all registered loans in a structured table.
- Shows loan details such as Outstanding Amount, EMI, Monthly Income, and Overdue Months.
- Provides action buttons including **Analyze**, **AI Letter**, **Negotiation Strategy**, **Edit**, and **Delete**.
- Confirms successful retrieval and management of loan records.



The screenshot shows the 'Loan Management' table in the FinRelief AI application. The table has a header row with the following columns: LOAN NAME, OUTSTANDING, EMI, MONTHLY INCOME, OVERDUE (MONTHS), and ACTIONS. The table contains one data row for the 'education loan'.

LOAN NAME	OUTSTANDING	EMI	MONTHLY INCOME	OVERDUE (MONTHS)	ACTIONS
education loan	₹26,000	₹15,000	₹30,000	2	Analyze Letter Strategy Edit Delete

At the top right of the table, there is a button labeled '+ Add Loan'.